

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282289

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

YASAKU; Shun et al.

SYSTEM FOR DISPLAYING IMAGE AROUND WORK VEHICLE, METHOD, AND WORK VEHICLE

Abstract

A monitor control device (**13**) selects any one of camera images captured by a camera system (**11**) based on a steering angle of front wheels (**41**) and outputs a signal for displaying the selected image on a display device.

Inventors: YASAKU; Shun (Tokyo, JP), TAMURA; Koichi (Tokyo, JP),
YAMAGUCHI; Kota (Tokyo, JP)

Applicant: KOMATSU LTD. (Tokyo, JP)

Family ID: 1000008673017

Appl. No.: 18/858371

**Filed (or PCT
Filed):** September 29, 2023

PCT No.: PCT/JP2023/035716

Foreign Application Priority Data

JP 2022-158040

Sep. 30, 2022

Publication Classification

Int. Cl.: B60R1/24 (20220101); B60R1/25 (20220101); B60R1/26 (20220101)

U.S. Cl.:

CPC B60R1/24 (20220101); B60R1/25 (20220101); B60R1/26 (20220101); B60R2300/105 (20130101); B60R2300/302 (20130101); B60R2300/802 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a system and a method for displaying a peripheral image of a work vehicle, and the work vehicle.

[0002] Priority is claimed on Japanese Patent Application No. 2022-158040, filed Sep. 30, 2022, the content of which is incorporated herein by reference.

BACKGROUND ART

[0003] Patent Document 1 discloses a technology related to a peripheral monitoring system of a work vehicle.

CITATION LIST

Patent Document

[0004] Patent Document 1: Japanese Unexamined Patent Application, First Publication No. 2022-127328

SUMMARY OF INVENTION

Technical Problem

[0005] The peripheral monitoring system of the work vehicle disclosed in Patent Document 1 includes a display unit that displays images of the periphery of the work vehicle. The operator of the work vehicle can recognize the situation of the periphery of the work vehicle by checking the image displayed on the display unit. Meanwhile, the cab of the work vehicle is disposed on the rear side of the front wheels. Therefore, when the operator allows the work vehicle to travel, it is difficult to recognize the situation in front of the work vehicle in a traveling direction according to steering operation. An object of the present disclosure is to provide a system and a method for displaying a peripheral image of a work vehicle, which can make an operator appropriately recognize a situation in front of the work vehicle in a traveling direction according to steering operation, and the work vehicle.

Solution to Problem

[0006] According to an aspect of the present invention, there is provided a system for displaying a peripheral image of a work vehicle including a vehicle body, and front wheels that are attached so as to be steerable to a front end part of the vehicle body, the system including: a steering angle sensor configured to detect a steering angle of the front wheel; a plurality of cameras that are provided on the vehicle body to image in different directions; a display device; and a processor. The processor selects any one of images captured by the plurality of cameras based on the detected steering angle of the front wheel. The processor outputs a signal for displaying the selected image on the display device.

[0007] According to an aspect of the present invention, there is provided a method for displaying a peripheral image of a work vehicle including a vehicle body, front wheels that are attached so as to be steerable to a front end part of the vehicle body, a steering angle sensor configured to detect a steering angle of the front wheel, a plurality of cameras that are provided on the vehicle body to image in different directions, and a display device, the method including: a step of detecting the steering angle of the front wheels detected by the steering angle sensor; a step of selecting any one of images captured by the plurality of cameras based on the steering angle of the front wheel; and a step of outputting a signal for displaying the selected image on the display device.

[0008] According to an aspect of the present invention, there is provided a work vehicle including: a vehicle body; front wheels that are attached so as to be steerable to a front end part of the vehicle body; a cab that is supported by the vehicle body to be disposed on a rear side of the front wheel; a steering angle sensor configured to detect a steering angle of the front wheel; a plurality of cameras that are provided on the vehicle body to image in different directions; a display device that is

provided in the cab; a processor. The processor selects any one of images captured by the plurality of cameras based on the detected steering angle of the front wheel. The processor outputs a signal for displaying the selected image on the display device.

Advantageous Effects of Invention

[0009] According to the above aspect, the work vehicle can make the operator appropriately recognize the situation in front of the work vehicle in the traveling direction according to the steering operation.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a perspective view of a work vehicle according to a first embodiment.

[0011] FIG. 2 is a side view of a work vehicle according to the first embodiment.

[0012] FIG. 3 is a top view schematically showing an articulation mechanism of a work vehicle according to the first embodiment.

[0013] FIG. 4 is a functional block diagram showing the configuration of a work vehicle according to the first embodiment.

[0014] FIG. 5 is a top view schematically showing a camera system according to the first embodiment.

[0015] FIG. 6 is a top view schematically showing a radar system according to the first embodiment.

[0016] FIG. 7 is a diagram showing the relationship between a steering angle and a selected camera image according to the first embodiment.

[0017] FIG. 8 is a diagram showing a display device according to the first embodiment.

[0018] FIG. 9 is a flowchart showing an image display method according to the first embodiment.

[0019] FIG. 10 is a block diagram showing a computer system according to the first embodiment.

DESCRIPTION OF EMBODIMENTS

[0020] In the present embodiment, a local coordinate system is set in a work vehicle 1, and a positional relationship of each of units will be described with reference to the local coordinate system. In the local coordinate system, a first axis extending in the left-right direction (the vehicle width direction) of the work vehicle 1 is defined as the X axis, a second axis extending in the front-rear direction of the work vehicle 1 is defined as the Y axis, and a third axis extending in the up-down direction of the work vehicle 1 is defined as the Z axis. The X axis and the Y axis are orthogonal to each other. The Y axis and the Z axis are orthogonal to each other. The Z axis and the X axis are orthogonal to each other. A +X direction is the right direction, and a -X direction is the left direction. A +Y direction is the front direction, and a -Y direction is the rear direction. A +Z direction is the up direction, and a -Z direction is the down direction.

First Embodiment

Work Vehicle

[0021] Hereinafter, the embodiment will be described in detail with reference to the drawings. FIG. 1 is a perspective view of a work vehicle 1 according to the first embodiment. FIG. 2 is a side view of a work vehicle 1 according to the first embodiment. The work vehicle 1 according to the first embodiment is a motor grader.

[0022] As shown in FIGS. 1 and 2, the work vehicle 1 includes a vehicle body 2, work equipment 3, wheels 4, and a cab 5.

[0023] The work vehicle 1 travels on the work site by the wheels 4. The work vehicle 1 performs works using the work equipment 3. Examples of the work performed by the work vehicle 1 include leveling work, road surface cutting work, excavation work, snow removal work, and material mixing work.

[0024] The vehicle body 2 supports the work equipment 3. The vehicle body 2 includes a front vehicle body 21 and a rear vehicle body 22. The front vehicle body 21 is disposed in front of the rear vehicle body 22. The front vehicle body 21 is swingably coupled to the rear vehicle body 22 via a coupling shaft 201. The front vehicle body 21 is swingable with respect to the rear vehicle body 22 in the left-right direction about the coupling shaft 201. The coupling shaft 201 is a shaft extending in the Z-axis direction. The front vehicle body 21 includes a rear end part 21r that is coupled to the rear vehicle body 22 and a front end part 21f that is provided on the side opposite to the rear end part 21r.

[0025] A counterweight 28 is attached to the front end part 21f of the front vehicle body 21. The counterweight 28 is a type of attachment attached to the front vehicle body 21. The counterweight 28 is mounted on the front vehicle body 21 to increase the downward load applied to front wheels 41, enabling steering and increasing the pressing load of the blade 32.

[0026] The rear vehicle body 22 extends rearward from the coupling shaft 201. The rear vehicle body 22 includes a front end part 22f that is coupled to the front vehicle body 21 and a rear end part 22r that is provided on the side opposite to the front end part 22f. A front end part 21f of the front vehicle body 21 is the front end part of the vehicle body 2. A rear end part 22r of the rear vehicle body 22 is the rear end part of the vehicle body 2.

[0027] FIG. 3 is a top view schematically showing an articulation mechanism 23 of the work vehicle 1 according to the first embodiment.

[0028] As shown in FIG. 3, the front vehicle body 21 and the rear vehicle body 22 are swingably coupled by the coupling shaft 201. The front vehicle body 21 is swung with respect to the rear vehicle body 22 by the articulation mechanism 23. The front vehicle body 21 is swung with respect to the rear vehicle body 22 by the articulation mechanism 23 so as to change the angle (articulation angle) formed with the rear vehicle body 22.

[0029] The articulation mechanism 23 includes an articulation cylinder 24 coupled to the front vehicle body 21 and the rear vehicle body 22. The articulation cylinder 24 expands and contracts, causing the front vehicle body 21 to swing in the left-right direction with respect to the rear vehicle body 22. The swing of the front vehicle body 21 with respect to the rear vehicle body 22 is performed by expanding and contracting the articulation cylinder 24 coupled between the front vehicle body 21 and the rear vehicle body 22 through the operation of the operation device 6. The articulation cylinder 24 is, for example, a hydraulic cylinder. An articulation angle sensor 25 is attached to the rear vehicle body 22 and detects the articulation angle, which is the swing angle of the front vehicle body 21 with respect to the rear vehicle body 22.

[0030] The work vehicle 1 swings (articulates) the front vehicle body 21 with respect to the rear vehicle body 22, thereby, reduces the swing radius during swinging and can perform ditch digging or slope cutting by offset traveling. The offset traveling refers to the straight travel of the work vehicle 1 by setting the swing direction of the front vehicle body 21 with respect to the rear vehicle body 22 and the steering directions of the front wheels 41 with respect to the front vehicle body 21 in opposite directions.

[0031] The rear vehicle body 22 supports a power source 27. The rear vehicle body 22 includes an exterior 26. The exterior 26 forms an engine room that houses the power source 27. The power source 27 is, for example, an engine.

[0032] The work equipment 3 is supported by the front vehicle body 21. The work equipment 3 includes a drawbar 30, a swing circle 31, a blade 32, a swing motor 33, and various cylinders 34 to 38.

[0033] The drawbar 30 is disposed below the front vehicle body 21. A front end part of the drawbar 30 is coupled to a front end part 21f of the front vehicle body 21 via a ball shaft portion. A front end part of the drawbar 30 is swingably supported by a front end part 21f of the front vehicle body 21. A rear end part of the drawbar 30 is supported by the front vehicle body 21 via a pair of lift cylinders 34 and 35. The drawbar 30 is suspended from the front vehicle body 21 via the pair of lift

cylinders **34** and **35**. A drawbar shift cylinder **36** is attached to a side end part of the front vehicle body **21** and the drawbar **30**. The drawbar **30** moves in the left-right direction with respect to the front vehicle body **21** by the expansion and contraction of the drawbar shift cylinder **36**.

[0034] The swing circle **31** is disposed below the front vehicle body **21**. The swing circle **31** is disposed below the drawbar **30**. The swing circle **31** is supported to be swingable at a rear end part of the drawbar **30**. The swing circle **31** can be driven to swing in a clockwise or counterclockwise direction with respect to the drawbar **30** as viewed from above the vehicle by the swing motor **33**. The swing drive of the swing circle **31** adjusts a tilt angle (blade propulsion angle) of the blade **32** with respect to the front vehicle body **21** in a plan view.

[0035] The blade **32** is disposed between the front end part **21f** of the front vehicle body **21** and the rear end part **22r** of the rear vehicle body **22**. The blade **32** is supported by the swing circle **31**. The blade **32** is supported by the drawbar **30** via the swing circle **31**. The blade **32** is supported by the front vehicle body **21** via the drawbar **30**. The swing circle **31** is swung to adjust a propulsion angle of the blade **32**. The propulsion angle of the blade **32** refers to the tilt angle of the blade **32** with respect to the Y axis.

[0036] The pair of lift cylinders **34** and **35** support the drawbar **30**. The pair of lift cylinders **34** and **35** support the blade **32** via the drawbar **30**. The rear end part of the drawbar **30** moves in the up-down direction with respect to the front vehicle body **21** by the synchronized expansion and contraction of the pair of lift cylinders **34** and **35**. That is, the heights of the drawbar **30** and the blade **32** are adjusted by the synchronized expansion and contraction of the pair of lift cylinders **34** and **35**. In addition, the drawbar **30** swings about an axis along the Y axis due to the different expansions and contractions of the lift cylinders **34** and **35**.

[0037] A blade shift cylinder **37** is attached to the swing circle **31** and the blade **32**, and is disposed along the longitudinal direction of the blade **32**. The blade **32** is moved in the left-right direction with respect to the swing circle **31** by the blade shift cylinder **37**.

[0038] The tilt cylinder **38** is attached to the swing circle **31** and the blade **32**. By expanding and contracting the tilt cylinder **38**, the blade **32** swings about an axis extending in the longitudinal direction of the blade **32** with respect to the swing circle **31**.

[0039] As described above, the blade **32** is configured to be able to change the tilt angle with respect to the front vehicle body **21**, move in the up-down direction with respect to the vehicle, swing about the axis along the Y axis, move in the left-right direction with respect to the swing circle **31**, and swing about the axis extending in the longitudinal direction of the blade **32**, via the drawbar **30** and the swing circle **31**.

[0040] A wheel **4** support a vehicle body **2**. The wheels **4** includes the front wheels **41** and rear wheels **42**. The front wheels **41** are disposed in front of the rear wheels **42**. In the front-rear direction, the blade **32** is disposed between the front wheels **41** and the rear wheels **42**. The front wheels **41** are disposed in front of the blade **32**. The rear wheels **42** are disposed on the rear side of the blade **32**. In the first embodiment, the work vehicle **1** is provided with traveling wheels consisting of two front wheels **41**, one on each side, and four rear wheels **42**, two on each side, making a total of six wheels. However, the number and arrangement of the front wheels **41** and the rear wheels **42** are not limited thereto.

[0041] The front wheels **41** are attached so as to be steerable to the front end part **21f** of the front vehicle body **21**. The front wheels **41** are attached to the front vehicle body **21** via a steering mechanism **43**. The front wheels **41** can change their direction with respect to the front vehicle body **21** by the steering mechanism **43**. The front wheels **41** operate to change an angle (steering angle) with respect to the front vehicle body **21**.

[0042] The rear wheels **42** are rotatably attached to the rear vehicle body **22**. The rear wheels **42** rotate based on the power generated by the power source **27**. The rear wheels **42** rotate based on the power generated by the power source **27** via the power transmission device **46**.

[0043] The steering mechanism **43** includes a steering cylinder **44**. The steering angle of the front

wheels **41** is changed by expanding and contracting the steering cylinder **44**. The change in the steering angle of the front wheels **41** is performed by expanding and contracting the steering cylinder **44** through the operation of the operation device **6**. The steering cylinder **44** is, for example, a hydraulic cylinder. A steering angle sensor **45** is attached to the steering mechanism **43** and detects the steering angle of the front wheels **41** with respect to the front vehicle body **21**. The steering angle sensor **45** may be, for example, a cylinder stroke sensor that detects the length of the steering cylinder **44**.

[0044] The power transmission device **46** transmits the power generated by the power source **27** to the rear wheels **42** by changing the torque, rotation speed, and rotation direction. The power transmission device **46** switches between forward movement and rearward movement of the work vehicle **1**. The work vehicle **1** switches between the forward movement and the rearward movement by the power transmission device **46**. The power transmission device **46** is not limited to a configuration that includes a plurality of gears and a plurality of clutches. The power transmission device **46** may have a configuration including a hydraulic pump and a hydraulic motor, and including a hydraulic static transmission (HST) or a hydraulic mechanical transmission (HMT), which converts the power generated by the power source **27** into a hydraulic pressure and transmits the hydraulic pressure. Alternatively, the power transmission device **46** may have a configuration including an electric mechanical transmission (EMT) with a generator and an electric motor instead of the hydraulic pump and the hydraulic motor.

[0045] The cab **5** is supported by the front vehicle body **21**. The cab **5** is disposed at the rear end part **21r** of the front vehicle body **21**. The cab **5** forms a space for the operator to board. The operation device **6**, a display device **15**, and various other operation devices are disposed inside the cab **5**.

[0046] FIG. **4** is a functional block diagram showing the configuration of the work vehicle **1** according to the first embodiment. As shown in FIG. **4**, the work vehicle **1** includes the operation device **6**, a vehicle control device **7**, and a peripheral monitoring system **10**.

[0047] The operation device **6** generates an operation signal for operating the work vehicle **1**. The operation device **6** is operated by the operator. The operation device **6** includes a vehicle speed operation device **61**, a forward/rearward movement switching device **62**, a braking operation device **63**, a steering operation device **64**, a work equipment operation device **65**, and a parking brake operation device **66**. When the operation device **6** is operated by the operator, the operation device **6** generates an operation signal. The operation signal generated by the operation device **6** is output to the vehicle control device **7** and the monitor control device **13**. The operation signal output by the operation device **6** includes an operation signal for traveling the work vehicle **1**. The operation device **6** outputs an operation signal based on the operator's operation to perform traveling, braking, and steering of the work vehicle **1**. The term "traveling" refers to the forward movement or the rearward movement of the work vehicle **1**. The braking refers to the deceleration or stopping of the work vehicle **1**. The steering refers to a change in the traveling direction of the work vehicle **1**.

[0048] The forward/rearward movement switching device **62** is, for example, a three-position alternate switch. The forward/rearward movement switching device **62** is operable in a forward position, a neutral position, and a rearward position. The operator operates the forward/rearward movement switching device **62** to output an operation signal for switching the traveling direction of the work vehicle **1** to forward, neutral, or rearward.

[0049] In the first embodiment, the steering operation device **64** is a steering wheel. The operator operates the steering operation device **64** to change the steering angle of the front wheels **41**, thereby changing the traveling direction of the work vehicle **1**. In addition, the steering operation device **64** is not limited to the steering wheel and may, for example, be a steering lever that is steered by lever operation. Alternatively, a configuration can be adopted in which both the steering wheel and the steering lever are provided.

[0050] An operation amount sensor **640** for detecting the operation amount of the steering operation device **64** by the operator is attached to the steering operation device **64**. In the first embodiment, the operation amount sensor **640** is a steering wheel sensor attached to the steering wheel. The operation amount sensor **640** detects the operation amount of the steering operation device **64** by the operator and outputs operation amount data to the vehicle control device **7**. In the first embodiment, the operation amount sensor **640** is, for example, a shaft displacement sensor that detects an angle displacement of a steering wheel shaft generated by the rotation of the steering wheel. In addition, in a case where the steering operation device **64** is the steering lever, the operation amount sensor **640** may be a position sensor that detects an angle position of the steering lever.

[0051] The parking brake operation device **66** is, for example, a toggle switch. The operator operates the parking brake operation device **66**, whereby an operation signal for switching the parking brake to the engaged state or the released state is output.

[0052] The vehicle control device **7** outputs a control command for operating the work vehicle **1** based on the operation signal from the operation device **6**.

[0053] The control command output from the vehicle control device **7** in order to perform the traveling of the work vehicle **1** includes a control command for increasing the output of the power source **27** and a control command for switching between the forward movement and the rearward movement of the work vehicle **1** by the power transmission device **46**. The control command for increasing the output of the power source **27** is output based on the operation signal output from the vehicle speed operation device **61**. The control command for switching between the forward movement and the rearward movement of the work vehicle **1** by the power transmission device **46** is output based on the operation signal output from the forward/rearward movement switching device **62**.

[0054] The control command output from the vehicle control device **7** in order to perform the braking of the work vehicle **1** includes a control command for operating a service brake (not shown). The control command for operating the service brake is output based on the operation signal output from the braking operation device **63**.

[0055] The control command output from the vehicle control device **7** in order to steer the work vehicle **1** includes a control command for operating the steering cylinder **44**. The control command for operating the steering cylinder **44** is output based on the operation signal output from the steering operation device **64**.

[0056] The control command output from the vehicle control device **7** in order to swing the front vehicle body **21** with respect to the rear vehicle body **22** includes a control command for operating the articulation cylinder **24**. The control command for operating the articulation cylinder **24** is output based on the operation signal output from the work equipment operation device **65**.

[0057] The control command output from the vehicle control device **7** in order to operate the parking brake (not shown) is output based on the operation signal output from the parking brake operation device **66**.

[0058] The vehicle control device **7** acquires articulation angle data detected by the articulation angle sensor **25**, steering angle data detected by the steering angle sensor **45**, and operation amount data detected by the operation amount sensor **640**. In the first embodiment, the vehicle control device **7** outputs the steering angle data to the monitor control device **13**, which will be described below. In addition, the steering angle data output by the vehicle control device **7** is not limited to data acquired from the steering angle sensor **45**. The steering angle data may be calculated based on, for example, the operation amount data detected by the operation amount sensor **640**.

[0059] In the steering angle data according to the first embodiment, a value of the steering angle when the front wheels **41** are parallel to the front-rear direction of the front vehicle body **21** is set to zero, a value of the steering angle when the front wheels **41** are directed to the right side with respect to the front vehicle body **21** in the front-rear direction is set to a positive number, and a

value of the steering angle when the front wheels **41** are directed to the left side with respect to the front vehicle body **21** in the front-rear direction is set to a negative number.

Peripheral Monitoring System **10**

[0060] The peripheral monitoring system **10** monitors the periphery of the work vehicle **1**. The peripheral monitoring system **10** includes a camera system **11**, a radar system **12**, a monitor control device **13**, an operation unit **14**, and a display device **15**.

[0061] The camera system **11** includes a plurality of cameras. The camera includes an optical system and an image sensor. Examples of the image sensor include a couple charged device (CCD) image sensor and a complementary metal-oxide-semiconductor (CMOS) image sensor. The camera is provided on the vehicle body **2** of the work vehicle **1** to image in different directions. The camera images the periphery of the work vehicle **1** to acquire an image of the periphery of the work vehicle **1**. The camera images at least the periphery of the vehicle body **2**. In the following description, the image acquired by the camera of the camera system **11** is appropriately referred to as a camera image.

[0062] The radar system **12** includes a plurality of radars. The radar is provided on the vehicle body **2** of the work vehicle **1**. The radar detects obstacles of the periphery of the work vehicle **1** in a non-contact manner.

[0063] FIG. **5** is a top view schematically showing the camera system **11** according to the first embodiment. As shown in FIGS. **1**, **2**, and **5**, the camera system **11** includes a first camera **111** provided on the front vehicle body **21**, a second camera **112** provided on the front vehicle body **21**, a third camera **113** provided on the rear vehicle body **22**, a fourth camera **114** provided on the rear vehicle body **22**, and a fifth camera **115** provided on the rear vehicle body **22**. The first camera **111** and the second camera **112** image in front of the work vehicle **1**. The third camera **113**, the fourth camera **114**, and the fifth camera **115** image of the rear side of the work vehicle **1**.

[0064] The first camera **111** is provided at the left part of the front end part **21f** of the front vehicle body **21**. The first camera **111** images the left front side of the work vehicle **1**. An imaging range **M1** of the first camera **111** is defined in the left front side of the work vehicle **1**.

[0065] The second camera **112** is provided at the right part of the front end part **21f** of the front vehicle body **21**. The second camera **112** images the right front side of the work vehicle **1**. An imaging range **M2** of the second camera **112** is defined in the right front side of the work vehicle **1**.

[0066] The third camera **113** is provided at the left part of the rear vehicle body **22**. The third camera **113** images the left side of the work vehicle **1**. An imaging range **M3** of the third camera **113** is defined in the left side of the work vehicle **1**.

[0067] The fourth camera **114** is provided at the right part of the rear vehicle body **22**. The fourth camera **114** images the right side of the work vehicle **1**. An imaging range **M4** of the fourth camera **114** is defined in the right side of the work vehicle **1**.

[0068] The fifth camera **115** is provided at the rear end part **22r** of the rear vehicle body **22**. The fifth camera **115** images the rear side of the work vehicle **1**. An imaging range **M5** of the fifth camera **115** is defined in the rear side of the work vehicle **1**.

[0069] In addition, the attachment position of the camera system **11** is not limited to the above-described position and is not particularly limited as long as the camera system **11** can acquire images of the periphery of the work vehicle **1**. For example, the first camera **111** and the second camera **112** may be provided on the counterweight **28**. For example, the third camera **113**, the fourth camera **114**, and the fifth camera **115** may be provided in the cab **5**.

[0070] At least a part of the imaging range **M1** and the imaging range **M2** overlap. At least a part of the imaging range **M3** and the imaging range **M5** overlap. At least a part of the imaging range **M4** and the imaging range **M5** overlap.

[0071] A non-imaging range **NR1**, in which an image is not captured by the camera system **11**, is provided at a part of the periphery of the cab **5**. The non-imaging range **NR1** is defined in both the left front side of the cab **5** and in the right front side of the cab **5**.

[0072] FIG. 6 is a top view schematically showing the radar system 12 according to the first embodiment. As shown in FIGS. 1, 2, and 6, the radar system 12 includes a first radar 121 provided on the front vehicle body 21, a second radar 122 provided on the front vehicle body 21, a third radar 123 provided on the rear vehicle body 22, a fourth radar 124 provided on the rear vehicle body 22, and a fifth radar 125 provided on the rear vehicle body 22.

[0073] The first radar 121 is provided at the left part of the front end part 21f of the front vehicle body 21. The first radar 121 detects the left front side of the work vehicle 1. A detection range D1 of the first radar 121 is defined in the left front side of the work vehicle 1.

[0074] The second radar 122 is provided at the right part of the front end part 21f of the front vehicle body 21. The second radar 122 detects the right front side of the work vehicle 1. A detection range D2 of the second radar 122 is defined in the right front side of the work vehicle 1.

[0075] The third radar 123 is provided at the left part of the rear vehicle body 22. The third radar 123 detects the left side of the work vehicle 1. A detection range D3 of the third radar 123 is defined in the left side of the work vehicle 1.

[0076] The fourth radar 124 is provided at the right part of the rear vehicle body 22. The fourth radar 124 detects the right side of the work vehicle 1. A detection range D4 of the fourth radar 124 is defined in the right side of the work vehicle 1.

[0077] The fifth radar 125 is provided at the rear end part 22r of the rear vehicle body 22. The fifth radar 125 detects the rear side of the work vehicle 1. A detection range D5 of the fifth radar 125 is defined in the rear side of the work vehicle 1.

[0078] In addition, the attachment position of the radar system 12 is not limited to the above-described position and is not particularly limited as long as it is a position where obstacles of the periphery of the work vehicle 1 can be detected. For example, the first radar 121 and the second radar 122 may be provided on the counterweight 28. For example, the third radar 123, the fourth radar 124, and the fifth radar 125 may be provided in the cab 5.

[0079] At least a part of the detection range D1 and the detection range D2 overlap. At least a part of the detection range D3 and the detection range D5 overlap. At least a part of the detection range D4 and the detection range D5 overlap.

[0080] A non-detection range NR2 that is not detected by the radar system 12 is provided in a part of the periphery of the cab 5. The non-detection range NR2 is defined in the left front side of the cab 5 and in the right front side of the cab 5.

[0081] The monitor control device 13 includes a computer system. The monitor control device 13 includes an acquisition unit 131, a first display image generation unit 132, a second display image generation unit 133, a boundary image generation unit 134, a symbol image generation unit 135, a determination unit 136, a selection unit 137, a storage unit 138, and a display control unit 139. The monitor control device 13 may include a single computer system or may function as the monitor control device 13 through the cooperation of multiple computer systems.

[0082] The acquisition unit 131 acquires a camera image from the camera system 11. The acquisition unit 131 acquires a first camera image showing the situation of the left front side of the front vehicle body 21 from the first camera 111. The acquisition unit 131 acquires the second camera image showing the situation of the right front side of the front vehicle body 21 from the second camera 112. The acquisition unit 131 acquires a third camera image showing the situation of the left side of the rear vehicle body 22 from the third camera 113. The acquisition unit 131 acquires the fourth camera image showing the situation of the right side of the rear vehicle body 22 from the fourth camera 114. The acquisition unit 131 acquires a fifth camera image showing the situation of the rear side of the rear vehicle body 22 from the fifth camera 115. The acquisition unit 131 acquires the radar detection data from the radar system 12.

[0083] The acquisition unit 131 acquires the steering angle data of the front wheels 41 from the vehicle control device 7. The acquisition unit 131 acquires an operation signal from the operation device 6. In the first embodiment, the acquisition unit 131 acquires the operation signal from the

forward/rearward movement switching device **62** and the operation signal from the parking brake operation device **66**.

[0084] The first display image generation unit **132** generates a first display image **IM1** showing a first part of the periphery of the work vehicle **1** based on the first camera image of the periphery of the work vehicle **1** captured by the first camera **111** and the second camera image of the periphery of the work vehicle **1** captured by the second camera **112**. In the first embodiment, the first display image **IM1** includes the front part of the periphery of the work vehicle **1**. The first display image **IM1** includes the periphery of the front vehicle body **21**.

[0085] The first display image **IM1** is a first viewpoint image viewed from a first viewpoint different from the position of the first camera **111** and the position of the second camera **112**. In the first embodiment, the first display image generation unit **132** generates a panoramic image showing the periphery of the front vehicle body **21** based on the first camera image and the second camera image.

[0086] The panoramic image refers to an image generated by combining a plurality of camera images acquired by each of the plurality of cameras with images in a plurality of visual line directions with a reference viewpoint for the generation of the panoramic image as the viewpoint position. For example, the panoramic image of the entire circumference is an image in which images corresponding to a plurality of visual line directions in the range of 0° to $+180^\circ$ and the range of 0° to -180° , with the reference viewpoint as the center, are continuously connected. In a case of a panoramic image of the entire circumference, an image in any visual line direction within a range of 360 degrees is obtained. In addition, the panoramic image need not correspond to the entire circumference. For example, an image corresponding to the visual line direction in a range of 0° to $+90^\circ$ and a range of 0° to -90° (a 180° panoramic image) may be used. In addition, the panoramic image may be an image in which images corresponding to a plurality of visual line directions obtained by rotating the visual line directions in the horizontal direction are continuously connected, or may be an image in which images corresponding to a plurality of visual line directions obtained by rotating the visual line directions in the vertical direction are continuously connected. In addition, the panoramic image may be an image in which images corresponding to a plurality of visual line directions obtained by changing the visual line direction in three dimensions are continuously connected, as in the all-sky image. In the first embodiment, the panoramic image is generated such that the right end part of the first camera image and the left end part of the second camera image are connected to each other. In the following description, the first display image **IM1** will be appropriately referred to as the panoramic image **IM1**.

[0087] The second display image generation unit **133** generates a second display image **IM2** showing a second part of the periphery of the work vehicle **1** based on the third camera image of the periphery of the work vehicle **1** captured by the third camera **113**, the fourth camera image of the periphery of the work vehicle **1** captured by the fourth camera **114**, and the fifth camera image of the periphery of the work vehicle **1** captured by the fifth camera **115**. In the first embodiment, the second display image **IM2** includes the rear part of the periphery of the work vehicle **1**. The second display image **IM2** includes the periphery of the rear vehicle body **22**.

[0088] The second display image **IM2** is a second viewpoint image viewed from a second viewpoint different from the positions of the third camera **113**, the fourth camera **114**, and the fifth camera **115**. In the first embodiment, the second display image generation unit **133** generates the bird's-eye view image showing the periphery of the rear vehicle body **22** based on the third camera image, the fourth camera image, and the fifth camera image. The bird's-eye view image refers to an image generated by converting a plurality of camera images acquired by each of the plurality of cameras into an upper viewpoint related to generation of the bird's-eye view image and combining the images.

[0089] The second display image **IM2** is an image with a display form different from that of the first display image **IM1**. For example, the second display image **IM2** is an image viewed from a

viewpoint different from that of the first display image IM1.

In the following description, the second display image IM2 will be appropriately referred to as a bird's-eye view image IM2.

[0090] The boundary image generation unit **134** generates a boundary image BI disposed between the panoramic image IM1 and the bird's-eye view image IM2. The boundary image BI refers to an image displayed at the boundary between the panoramic image IM1 and the bird's-eye view image IM2 to clearly distinguish between the panoramic image IM1 and the bird's-eye view image IM2.

[0091] The symbol image generation unit **135** generates a symbol image SI indicating the work vehicle **1**. In the first embodiment, the symbol image generation unit **135** changes the symbol image SI based on the operation signal from the operation device **6**. The symbol image SI refers to an image that simulates the work vehicle **1** as viewed from above.

[0092] The determination unit **136** determines the traveling state of the work vehicle **1** based on the operation signal from the operation device **6** acquired by the acquisition unit **131**. In the first embodiment, the determination unit **136** determines whether the work vehicle **1** has transitioned to the forward moving state or the neutral state based on the operation signal from the forward/rearward movement switching device **62** acquired by the acquisition unit **131** and the operation signal from the parking brake operation device **66** acquired by the acquisition unit **131**. Specifically, the determination unit **136** determines that the work vehicle **1** has transitioned to the forward moving state or the neutral state when the operation signal from the forward/rearward movement switching device **62** acquired by the acquisition unit **131** indicates the forward moving operation or the neutral operation of the power transmission device **46**, and the operation signal from the parking brake operation device **66** acquired by the acquisition unit **131** indicates the release operation of the parking brake.

[0093] In a case where the determination unit **136** determines that the work vehicle **1** has transitioned to the forward moving state or the neutral state, the selection unit **137** selects any one of the camera images captured by the camera of the camera system **11** based on the steering angle data of the front wheels **41** acquired by the acquisition unit **131**. In the first embodiment, when the determination unit **136** determines that the work vehicle **1** has transitioned to the forward moving state or the neutral state, the selection unit **137** selects either the camera image of the first camera **111** or the camera image of the second camera **112** based on the steering angle data of the front wheels **41** acquired by the acquisition unit **131**. The selection unit **137** selects either the camera image of the first camera **111** or the camera image of the second camera **112** by comparing the steering angle data of the front wheels **41** acquired by the acquisition unit **131** with the threshold value.

[0094] FIG. 7 is a diagram showing the relationship between the steering angle and the selected camera image according to the first embodiment. As shown in FIG. 7, when the steering angle is less than a first threshold value $th1$, the selection unit **137** selects the camera image of the first camera **111**. In a case where the steering angle is more than the second threshold value $th2$, the selection unit **137** selects the camera image of the second camera **112**. The first threshold value is smaller than the second threshold value. That is, the selection unit **137** according to the first embodiment provides hysteresis with respect to the threshold value of the steering angle.

[0095] The first threshold value $th1$ and the second threshold value $th2$ are set to angles different from the steering angle during straight moving. As a result, the selection unit **137** selects either the camera image of the first camera **111** or the camera image of the second camera **112** so that the operator can appropriately recognize the situation in the front side even when the work vehicle **1** is traveling straight. Both the first threshold value $th1$ and the second threshold value $th2$ according to the first embodiment are positive numbers. That is, the selection unit **137** selects the first camera **111** that images the left side not only when the front wheels are facing the left side but also when the front wheels are facing the front side and when the front wheels are facing slightly to the right side. For example, when the work vehicle **1** performs leveling work while traveling on the left side

of the road, the work vehicle **1** needs to travel close to the shoulder of the road on the left side of the road to dump the earth generated by the leveling work onto the shoulder of the road. Therefore, the selection unit **137** always selects the first camera **111** except when making a right turn, so that the operator can appropriately recognize the situation in the left front side. In addition, in a case where the operator wants to appropriately recognize the situation in the right front side, the first threshold value **th1** and the second threshold value **th2** may both be negative numbers. In addition, in another embodiment, the median value of the first threshold value **th1** and the second threshold value **th2** may be zero. In addition, the first threshold value **th1** and the second threshold value **th2** may be changed to desired values by the operation of the operator.

[0096] The storage unit **138** stores selection data indicating which camera image was last selected by the selection unit **137**.

[0097] The display control unit **139** generates a composite image **CI** by combining the panoramic image **IM1** generated by the first display image generation unit **132**, the bird's-eye view image **IM2** generated by the second display image generation unit **133**, the boundary image **BI** generated by the boundary image generation unit **134**, and the symbol image **SI** generated by the symbol image generation unit **135**. The display control unit **139** displays the generated composite image **CI** on the display unit **151** of the display device **15**. The display control unit **139** causes the display unit **151** of the display device **15** to display the single camera image **IS** showing the front of the work vehicle **1**. The single camera image **IS** is a camera image captured by the camera selected by the selection unit **137**.

[0098] The operation unit **14** includes a plurality of switches disposed in the cab **5**. A specific function is assigned to each of the plurality of switches. In a case where the switch is operated by the operator, an operation signal for a specific function is generated.

Display Device

[0099] FIG. **8** is a diagram showing the display device **15** according to the first embodiment. The display device **15** displays the periphery of the work vehicle **1**. The display device **15** displays at least the periphery of the vehicle body **2**. As shown in FIG. **8**, the display device **15** includes the display unit **151**. The display unit **151** may be, for example, a touch panel. In this case, the display unit **151** can function as the operation unit **14**. In addition, the operator may change the first threshold value **th1** and the second threshold value **th2** to desired values by performing a touch operation on the touch panel.

[0100] The display unit **151** includes a first region **151A** and a second region **151B**. The first region **151A** of the display unit **151** displays the single camera image **IS**. The second region **151B** of the display unit **151** displays the composite image **CI**. In the first embodiment, the second region **151B** is defined in the left side of the first region **151A**.

[0101] In the example shown in FIG. **8**, the single camera image **IS** displayed in the first region **151A** is the first camera image showing the situation of the left front side of the work vehicle **1** acquired by the first camera **111**.

[0102] In the example shown in FIG. **8**, the composite image **CI** displayed in the second region **151B** includes the panoramic image **IM1**, the bird's-eye view image **IM2**, the boundary image **BI**, and the symbol image **SI**. The panoramic image **IM1** is displayed on a part of the composite image **CI**. The bird's-eye view image **IM2** is displayed on a part of the composite image **CI**. A boundary image **BI** is displayed at the boundary between the panoramic image **IM1** and the second display image **IM2**. The symbol image **SI** is displayed on the central region of the composite image **CI**. The composite image **CI** is displayed such that the panoramic image **IM1**, the bird's-eye view image **IM2**, and the boundary image **BI** are arranged around the symbol image **SI**. The panoramic image **IM1** is disposed on the upper part of the composite image **CI**. The bird's-eye view image **IM2** is disposed on the lower part of the composite image **CI**. The boundary image **BI** is displayed on the side part of the symbol image **SI**. The composite image **CI** clarifies the positional relationship between the work vehicle **1** and the periphery of the work vehicle **1** through the symbol image **SI**.

[0103] In the first embodiment, the boundary image BI is displayed on the side part of the articulation mechanism **23S** of the symbol image SI. The boundary image BI is arranged to extend in the left-right direction in the composite image CI. The boundary image BI is displayed in a band shape on both the left side and the right side of the symbol image SI. The boundary image BI may be displayed on the display unit **151** between the front wheels and the cab of the symbol image SI. The boundary image BI may be displayed on a predetermined region below the panoramic image IM1 displayed as the composite image CI on the display unit **151**. The boundary image BI may be displayed on a predetermined region above the bird's-eye view image IM2 displayed in the composite image CI on the display unit **151**. The boundary image BI may be displayed on a predetermined region of the composite image CI on the display unit **151**. The predetermined region may be defined by an image coordinate system with the pixel at the upper left of the composite image CI as the origin.

[0104] As described above, a non-imaging range NR1, which is not imaged by the camera system **11**, is provided in a part of the periphery of the cab **5**. The panoramic image IM1 and the bird's-eye view image IM2 are not generated for the non-imaging range NR1. That is, in the display unit **151**, the region corresponding to the non-imaging range NR1 is a non-display region where the panoramic image IM1 and the bird's-eye view image IM2 are not displayed. The boundary image BI is generated to cover a non-display region where the panoramic image IM1 and the bird's-eye view image IM2 are not displayed on the display unit **151**.

[0105] A reference line GD2 is displayed in the second region **151B** in conjunction with the composite image CI. Similarly to a reference line GD1, the reference line GD2 indicates a reference for the distance from the vehicle body **2**. In the first embodiment, the reference line GD2 is displayed to be superimposed on the bird's-eye view image IM2. The reference line GD2 is disposed around the symbol image SI in the bird's-eye view image IM2. The reference line GD2 does not need to be disposed.

[0106] In a case where an obstacle OB is detected by the radar system **12**, the display control unit **139** may display a marker Mk on the display unit **151** to overlap the marker Mk with the obstacle OB imaged on the bird's-eye view image IM2. The marker MK functions as a symbol image for emphasizing the obstacle OB on the display unit **151**.

Image Display Method

[0107] FIG. **9** is a flowchart showing an image display method according to the first embodiment. When the key of the work vehicle **1** is turned on, the peripheral monitoring system **10** is activated.

[0108] The camera system **11** images the periphery of a motor grader **1**. The acquisition unit **131** acquires the camera image from the camera system **11** (step S1).

[0109] The first display image generation unit **132** generates a panoramic image IM1 showing the front part of the periphery of the work vehicle **1** based on the first camera image of the periphery of the work vehicle **1** captured by the first camera **111** and the second camera image of the periphery of the work vehicle **1** captured by the second camera **112** (step S2).

[0110] The second display image generation unit **133** generates the bird's-eye view image IM2 showing the rear part of the periphery of the work vehicle **1** based on the third camera image in the periphery of the work vehicle **1** captured by the third camera **113**, the fourth camera image of the periphery of the work vehicle **1** captured by the fourth camera **114**, and the fifth camera image of the periphery of the work vehicle **1** captured by the fifth camera **115** (step S3).

[0111] The boundary image generation unit **134** generates the boundary image BI (step S4).

[0112] The symbol image generation unit **135** generates the symbol image SI based on the operation signal of the operation device **6** (step S5).

[0113] The display control unit **139** combines the symbol image SI, the panoramic image IM1, the bird's-eye view image IM2, and the boundary image BI to generate a composite image CI (step S6).

[0114] The determination unit **136** determines whether the work vehicle **1** has transitioned to the forward moving state or the neutral state based on the operation signal from the forward/rearward

movement switching device **62** acquired by the acquisition unit **131** and the operation signal from the parking brake operation device **66** acquired by the acquisition unit **131** (step S7).

[0115] In a case where the determination unit **136** determines that the work vehicle **1** has transitioned to the forward moving state or the neutral state (step S7: YES), the acquisition unit **131** acquires the steering angle data from the vehicle control device **7** (step S8). Then, the selection unit **137** refers to the selection data stored in the storage unit **138** to determine whether the camera image last selected is the camera image of the first camera **111** (step S9).

[0116] On the other hand, in a case where the determination unit **136** determines that the work vehicle **1** has not transitioned to the forward moving state or the neutral state (step S7: NO), the process in FIG. **9** is ended.

[0117] In a case where it is determined that the camera image selected last time is the camera image of the first camera **111** (step S9: YES), the selection unit **137** determines whether the steering angle is more than the second threshold value $th2$ based on the steering angle data acquired in step S8 (step S10).

[0118] In a case where the determination is made that the steering angle is more than the second threshold value $th2$ (step S10: YES), the selection unit **137** selects the camera image of the second camera **112** (step S12).

[0119] In a case where the determination is made that the steering angle is not more than the second threshold value $th2$ (step S10: NO), the selection unit **137** selects the camera image of the first camera **111** (step S13).

[0120] In a case where the determination is made that the camera image selected last time is the camera image of the second camera **112** (step S9: NO), the selection unit **137** determines whether the steering angle is less than the first threshold value $th1$ based on the steering angle data acquired in step S8 (step S11).

[0121] In a case where the determination is made that the steering angle is less than the first threshold value $th1$ (step S11: YES), the selection unit **137** selects the camera image of the first camera **111** (step S13).

[0122] In a case where the determination is made that the steering angle is not less than the first threshold value $th1$ (step S11: NO), the selection unit **137** selects the camera image of the second camera **112** (step S12).

[0123] The storage unit **138** updates the selection data indicating which camera image is selected by the selection unit **137** (step S14).

[0124] The display control unit **139** outputs a display signal for displaying the composite image CI generated in step S6 and a display signal for displaying the camera image captured by the camera selected in step S12 or step S13 as the single camera image IS to the display device **15** (step S15).

Computer System

[0125] FIG. **10** is a block diagram showing a computer system **1000** according to the first embodiment. Each of the monitor control device **13** and the vehicle control device **7** includes the computer system **1000**. The computer system **1000** includes a processor **1001** such as a central processing unit (CPU), a main memory **1002** including a non-volatile memory such as a read only memory (ROM) and a volatile memory such as a random access memory (RAM), a storage **1003**, and an interface **1004** including an input/output circuit. For example, the function of the monitor control device **13** is stored in the storage **1003** as a computer program. The processor **1001** reads the computer program from the storage **1003** and loads it into the main memory **1002** to execute the above-described processing according to the computer program. The computer program may be distributed to the computer system **1000** via a network.

Operations and Effects

[0126] As described above, according to the first embodiment, the monitor control device **13** selects and displays the camera image related to the single camera image IS based on the steering angle of the front wheels **41**. As a result, the work vehicle **1** can appropriately recognize the

situation in front of the work vehicle **1** in the traveling direction according to the steering operation by the operator. In the work vehicle **1** including the front wheels **41** that are provided so as to be steerable at the front end part of the vehicle body **2** and the cab **5** that is supported by the vehicle body **2** to be disposed on the rear side of the front wheels **41**, the state of the vicinity of the front wheels **41**, which is difficult for an operator to recognize, can be notified to the operator. In particular, when the operator allows the work vehicle **1** to travel, the operator can appropriately recognize the situation in front of the work vehicle **1** in the traveling direction according to the steering operation.

[0127] In addition, in the first embodiment, the first threshold value $th1$ and the second threshold value $th2$ of the steering angle, at which the camera image captured by the first camera **111** and the camera image captured by the second camera **112** are switched, are set to angles different from the steering angle during straight moving. As a result, even when the work vehicle **1** travels straight, the operator can appropriately recognize the situation in the front side.

Second Embodiment

[0128] The peripheral monitoring system **10** according to the first embodiment selects the camera image related to the single camera image IS based on the steering angle. On the other hand, the peripheral monitoring system **10** according to the second embodiment further selects the camera with reference to the articulation angle of the front vehicle body **21**.

[0129] Specifically, the vehicle control device **7** calculates the sum of the articulation angle and the steering angle from the steering angle data detected by the steering angle sensor **45** and the articulation angle data detected by the articulation angle sensor **25**. Then, the selection unit **137** selects the camera image related to the single camera image IS based on the sum of the steering angle and the articulation angle calculated by the vehicle control device **7**. Specifically, the selection unit **137** selects either the camera image captured by the first camera **111** or the camera image captured by the second camera **112** by comparing the sum of the articulation angle and the steering angle with the first threshold value $th1$ or the second threshold value $th2$. For example, in a case where the articulation direction and the steering direction are opposite to each other, the articulation angle and the steering angle cancel each other out. Therefore, by selecting the camera image related to the single camera image IS using the sum of the articulation angle and the steering angle, the operator can more appropriately recognize the situation in the direction according to the turning.

[0130] In another embodiment, in a case where the front wheels **41** has the leaning function, the selection unit **137** may select the camera image related to the single camera image IS based on the leaning angle in addition to the articulation angle and the steering angle. The leaning function is a function of changing the turning angle of the work vehicle **1** by tilting the rotation axis of the front wheels **41** in the up-down direction.

Another Embodiment

[0131] The peripheral monitoring system **10** according to another embodiment may not include the display device **15** and the radar system **12**. For example, the peripheral monitoring system **10** according to another embodiment need only output a signal for displaying on the monitor. In addition, the peripheral monitoring system **10** according to another embodiment may detect obstacles by performing image analysis on the camera images from the camera system **11** instead of the radar system **12**. In addition, the peripheral monitoring system **10** according to another embodiment may not have a function of detecting obstacles. In addition, some components of the peripheral monitoring system **10** may be mounted inside the work vehicle **1**, while other components may be provided outside the work vehicle **1**. For example, the display device **15** of the peripheral monitoring system **10** may be disposed in a remote operation room located at a remote location from the work vehicle **1**.

[0132] The display control unit **139** according to the first embodiment causes the display device **15** to display the composite image CI and the single camera image IS, but is not limited thereto. For

example, the display control unit **139** according to another embodiment may display only the single camera image **IS** on the display device **15**.

[0133] The camera system **11** according to the first embodiment includes the first camera **111** and the second camera **112** provided on the front vehicle body **21**, but is not limited thereto. For example, the camera system **11** according to another embodiment may have any number of cameras, three or more, in the front vehicle body **21**.

[0134] The camera system **11** according to the first embodiment includes the third camera **113**, the fourth camera **114**, and the fifth camera **115** provided on the rear vehicle body **22**, but is not limited thereto. For example, the camera system **11** according to another embodiment may include two cameras or any number of cameras, which are four or more, provided on the rear vehicle body **22**.

[0135] In another embodiment, the imaging range **M1** and the imaging range **M2** may not overlap. The imaging range **M3** and the imaging range **M5** may not overlap. The imaging range **M4** and the imaging range **M5** may not overlap.

[0136] In another embodiment, the cab **5** may be supported by the rear vehicle body **22**.

[0137] In another embodiment, the work vehicle **1** may be an articulated dump truck having an articulation mechanism. In addition, the work vehicle **1** may be a wheel loader including an articulation mechanism and work equipment.

Industrial Applicability

[0138] According to the above aspect, the work vehicle can make the operator appropriately recognize the situation in front of the work vehicle in the traveling direction according to the steering operation.

Reference Signs List

[0139] **1**: Work vehicle [0140] **2**: Vehicle body [0141] **3**: Work equipment [0142] **4**: Wheel [0143] **5**: Cab [0144] **6**: Operation device [0145] **7**: Vehicle control device [0146] **10**: Peripheral monitoring system [0147] **11**: Camera system [0148] **12**: Radar system [0149] **13**: Monitor control device [0150] **14**: Operation unit [0151] **15**: Display device [0152] **21**: Front vehicle body [0153] **22**: Rear vehicle body [0154] **23**: Articulation mechanism [0155] **24**: Articulation cylinder [0156] **25**: Articulation angle sensor [0157] **26**: Exterior [0158] **27**: Power source [0159] **28**: Counterweight [0160] **30**: Drawbar [0161] **31**: Swing circle [0162] **32**: Blade [0163] **34**: Lift cylinder [0164] **35**: Lift cylinder [0165] **41**: Front wheel [0166] **42**: Rear wheel [0167] **43**: Steering mechanism [0168] **44**: Steering cylinder [0169] **45**: Steering angle sensor [0170] **46**: Power transmission device [0171] **61**: Vehicle speed operation device [0172] **62**: Forward/rearward movement switching device [0173] **63**: Braking operation device [0174] **64**: Steering operation device [0175] **65**: Work equipment operation device [0176] **66**: Parking brake operation device [0177] **111**: First camera [0178] **112**: Second camera [0179] **113**: Third camera [0180] **114**: Fourth camera [0181] **115**: Fifth camera [0182] **121**: First radar [0183] **122**: Second radar [0184] **123**: Third radar [0185] **124**: Fourth radar [0186] **125**: Fifth radar [0187] **131**: Acquisition unit [0188] **132**: First display image generation unit [0189] **133**: Second display image generation unit [0190] **134**: Boundary image generation unit [0191] **135**: Symbol image generation unit [0192] **136**: Determination unit [0193] **137**: Selection unit [0194] **138**: Storage unit [0195] **139**: Display control unit [0196] **151**: Display unit [0197] **201**: Coupling shaft [0198] **640**: Operation amount sensor

Claims

1. A system for displaying a peripheral image of a work vehicle including a vehicle body, and front wheels that are attached so as to be steerable to a front end part of the vehicle body, the system comprising: a steering angle sensor configured to detect a steering angle of the front wheel; a plurality of cameras that are provided on the vehicle body to image in different directions; a display device; and a processor, wherein the processor selects any one of images captured by the plurality of cameras based on the steering angle of the front wheel, and outputs a signal for displaying the

selected image on the display device.

2. The system according to claim 1, wherein the plurality of cameras include a first camera that is provided at the front end part of the vehicle body and images a left front side of the work vehicle, and a second camera that is provided at the front end part of the vehicle body and images a right front side of the work vehicle, and the processor selects either one of an image captured by the first camera or an image captured by the second camera based on the detected steering angle of the front wheel.

3. The system according to claim 2, wherein the processor selects either one of the image captured by the first camera or the image captured by the second camera, by comparing the steering angle of the front wheels with a threshold value, and the threshold value is different from a steering angle of the work vehicle during straight moving.

4. The system according to claim 1, wherein the work vehicle includes a power transmission device configured to switch between forward movement and rearward movement, and the processor, in a case where a command signal to switch the power transmission device to forward movement is received, selects any one of the images captured by the plurality of cameras based on the steering angle of the front wheel.

5. The system according to claim 1, wherein the work vehicle includes a parking brake, and in a case where a command signal for releasing the parking brake is received, the processor selects any one of the images captured by the plurality of cameras based on the steering angle of the front wheel.

6. The system according to claim 1, wherein the vehicle body of the work vehicle includes a rear vehicle body and a front vehicle body that is swingably coupled to the rear vehicle body, the system further comprises an articulation angle sensor configured to detect an articulation angle of the front vehicle body with respect to the rear vehicle body, and the processor selects any one of the images captured by the plurality of cameras based on a sum of the steering angle and the articulation angle.

7. A method for displaying a peripheral image of a work vehicle including a vehicle body, front wheels that are attached so as to be steerable to a front end part of the vehicle body, a steering angle sensor configured to detect a steering angle of the front wheel, a plurality of cameras that are provided on the vehicle body to image in different directions, and a display device, the method comprising the steps of: acquiring the steering angle of the front wheels detected by the steering angle sensor; selecting any one of images captured by the plurality of cameras based on the steering angle of the front wheel; and outputting a signal for displaying the selected image on the display device.

8. A work vehicle comprising: a vehicle body; front wheels that are attached so as to be steerable to a front end part of the vehicle body; a cab that is supported by the vehicle body to be disposed on a rear side of the front wheel; a steering angle sensor configured to detect a steering angle of the front wheel; a plurality of cameras that are provided on the vehicle body to image in different directions; a display device that is provided in the cab; and a processor, wherein the processor selects any one of images captured by the plurality of cameras based on the steering angle of the front wheel, and outputs a signal for displaying the selected image on the display device.
